

Table of Content

Advanced LMS Platform – Functional Requirements Document (FRD)

1. Project Overview
2. User Roles
 - 2.1 Admin
 - 2.2 Trainer / Lecturer
 - 2.3 Student / Learner
 - 2.4 Institution / Corporate Client
3. Core Functional Modules
 - 3.1 User Authentication & Access Control
 - Features
 - Advanced
 - 3.2 Course Management Module
 - Trainer Features
 - Course Types
 - 3.3 Paid Market Training Module
 - Features
 - Functionalities
 - Pricing Types
 - 3.4 Platform Access Pricing
 - Minimum Cost Access
 - Access Includes
 - 3.5 Live Training Management
 - Features
 - Student Features
 - Trainer Features
 - 3.6 Individual & Group Classes
 - Individual Sessions
 - Group Sessions
 - 3.7 Smart Classroom Module (IoT Enabled)
 - Scenario
 - Features
 - Classroom Hardware Integration
 - Functional Requirements
 - Remote Lecturer Teaching
 - Student Doubt Interaction
 - Motion Camera Tracking
 - Voice Detection AI
 - 3.8 Classroom Seating & Physical Room Management
 - Features

3.9 Container Classrooms / Mobile Classrooms (Phase - 2)

- Features

3.10 Free Online Training Library

- Features

- Search

3.11 Recording & Storage Module

- Features

- Storage

3.12 Common LMS Features

- Learning Progress

- Assessments

- Assignments

- Certificates

- Discussion Forum

- Notifications

3.13 Payment Module

- Features

- Gateways

3.14 Reports & Analytics

- Admin Dashboard

- Trainer Dashboard

3.15 AI Advanced Features

- AI Suggestions

- AI Classroom

4. Non Functional Requirements

- Performance

- Security

- Scalability

Advanced LMS Platform – Functional Requirements Document (FRD)

Project Name: Smart Hybrid LMS Platform

Version: 1.0

Prepared For: Development Team

Date: 17 April 2026

1. Project Overview

The LMS Platform is a **smart hybrid learning ecosystem** that supports:

- Self-paced online learning
 - Paid market training programs
 - Live virtual classes
 - Group and individual training
 - In-person classroom learning
 - Smart IoT-enabled classrooms
 - Remote faculty teaching physical classrooms
 - Recorded session storage
 - Free and paid training libraries
 - Container/mobile classrooms(Phase - 2)
-

2. User Roles

The system should support the following user roles:

2.1 Admin

- Platform configuration
- User management
- Trainer onboarding

- Pricing control
 - Course approval
 - Subscription management
 - Reports & analytics
 - Classroom management
 - Device management (IoT cameras/mics)
-

2.2 Trainer / Lecturer

- Create courses
 - Conduct live classes
 - Schedule sessions
 - Upload recordings
 - Manage attendance
 - Assign quizzes / assignments
 - Conduct one-on-one sessions
 - Group classes
 - Remote smart classroom teaching
-

2.3 Student / Learner

- Register/login
 - Browse courses
 - Purchase course
 - Access free library
 - Attend live classes
 - Join physical classes
 - Raise doubts
 - Track progress
 - Download certificates
-

2.4 Institution / Corporate Client

- Bulk student enrollments
- Group training requests
- Training analytics

- Custom batches
 - Physical classroom booking
-

3. Core Functional Modules

3.1 User Authentication & Access Control

Features

- Email login
- Mobile OTP login
- Social login
- SSO login
- Role-based access
- Multi-device login
- Session timeout
- Password reset

Advanced

- Biometric login (mobile app)
 - Device recognition
 - Login activity logs
-

3.2 Course Management Module

Trainer Features

- Create new course
- Course title
- Description

- Tags/category
- Skill level
- Duration
- Pricing
- Discount offers
- Thumbnail/video intro
- Course prerequisites

Course Types

- Self-paced online course
 - Live training batch
 - Physical classroom course
 - Hybrid course
 - Individual coaching
 - Group coaching
-

3.3 Paid Market Training Module

This is one of the most important modules from your requirement.

Features

create market-driven paid training programs.

Example:

- AI training
- Salesforce training
- Full stack training
- Interview prep
- Career mentorship

Functionalities

- Trainer sets course price
- Platform service charge
- Taxes/GST
- Revenue sharing

- Discount coupons
- Promo campaigns
- Referral credits

Pricing Types

- One-time payment
 - Subscription
 - Installments / EMI
 - Pay per session
 - Corporate pricing
 - Group pricing
-

3.4 Platform Access Pricing

Minimum Cost Access

Students should be able to pay **minimum platform access fee**.

Example:

- Monthly access: ₹xxx
- Quarterly: ₹xxx
- Annual: ₹xxxx

Access Includes

- Free library
 - Recorded classes
 - Basic courses
 - Community support
-

3.5 Live Training Management

This is the **live classroom module**.

Features

- Schedule live class
- Date/time
- Duration
- Trainer
- Batch size
- Registration limit
- Waiting list
- Auto reminders

Student Features

- Register for session
- Join live class
- Ask questions
- Chat box
- Raise hand
- Attendance tracking

Trainer Features

- Screen share
 - Whiteboard
 - Polls
 - Quiz
 - Breakout rooms
 - Recording
-

3.6 Individual & Group Classes

Individual Sessions

- 1:1 trainer booking
- Calendar scheduling
- Time slot booking
- Payment per hour

Group Sessions

- Multiple students
 - Topic-specific batch
 - Invite-based enrollment
 - Capacity controls
-

3.7 Smart Classroom Module (IoT Enabled)

This is your advanced smart classroom requirement.

Scenario

Students are physically present in classroom.

Faculty is remote.

The faculty takes class virtually.

Students interact through smart devices.

Features

Classroom Hardware Integration

- IoT microphone devices
 - Smart speakers
 - PTZ motion cameras
 - Face tracking camera
 - Smart board
 - Conference display screen
 - Auto voice amplification
-

Functional Requirements

Remote Lecturer Teaching

- Faculty joins remotely
 - Live video projected in classroom
 - Lecture visible on smart display
 - Shared content visible
-

Student Doubt Interaction

Each student seat should support:

- Desk mic / wireless mic
- Push-to-talk button
- Voice capture

When student asks doubt:

- mic activates
 - camera auto focuses
 - lecturer sees student live
-

Motion Camera Tracking

AI-enabled motion cameras should:

- detect speaking student
- auto zoom face
- focus active speaker
- switch screen automatically

This is critical.

Example:

When student in row 3 asks question,
camera captures face and displays on trainer screen.

Voice Detection AI

- detect active voice source
 - eliminate background noise
 - smart voice enhancement
-

3.8 Classroom Seating & Physical Room Management

Features

- Classroom capacity
- Seat allocation
- Attendance by seat
- Classroom layout map
- Device mapping per seat

Example:

Seat A1 → Mic ID 001

Seat A2 → Mic ID 002

3.9 Container Classrooms / Mobile Classrooms (Phase - 2)

This is a highly innovative feature.

Container classrooms are **portable learning spaces**.

Features

- mobile training units
- location mapping
- batch scheduling
- seat capacity
- power/device status
- connectivity monitoring

Useful for:

- rural education
 - corporate field training
 - mobile labs
-

3.10 Free Online Training Library

Features

- free video library
- downloadable notes
- ebooks
- recorded webinars
- sample lessons
- public certification resources

Search

- keyword search
 - category filters
 - trainer filters
 - popularity ranking
-

3.11 Recording & Storage Module

All recordings should be stored.

This is mandatory.

Features

- automatic live session recording
- session playback
- chapter markers

- transcript generation
- searchable timestamps

Storage

- cloud video storage
 - CDN delivery
 - backup retention
-

3.12 Common LMS Features

Learning Progress

- module completion %
 - video watch %
 - time spent
 - progress bar
 - resume learning
-

Assessments

- quizzes
 - MCQs
 - coding tests
 - descriptive answers
 - automated scoring
-

Assignments

- upload assignment
- trainer review
- feedback comments
- grading rubric

Certificates

- auto certificate generation
- completion certificate
- QR verification
- downloadable PDF

Discussion Forum

- course discussion boards
- trainer Q&A
- peer learning

Notifications

- email alerts
 - SMS
 - WhatsApp
 - push notifications
-

3.13 Payment Module

Features

- course payment
- platform subscription
- refunds
- coupon codes
- invoice generation
- GST invoice

Gateways

- Razorpay
 - Stripe
 - PayPal
 - UPI
-

3.14 Reports & Analytics

Admin Dashboard

- active users
- enrollments
- revenue
- trainer payouts
- attendance reports

Trainer Dashboard

- class performance
 - student engagement
 - doubt frequency
 - completion %
-

3.15 AI Advanced Features

Highly recommended.

AI Suggestions

- personalized course recommendations
- smart revision reminders
- performance alerts

AI Classroom

- speech-to-text notes
 - auto lecture summary
 - doubt clustering
 - engagement heatmap
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4. Non Functional Requirements

Performance

- 10,000 concurrent users
- 500 live classes simultaneously

Security

- encrypted video streams
- secure payments
- GDPR compliant
- role-based permissions

Scalability

- multi-tenant architecture
 - cloud native deployment
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